

Myopia: Current Concepts and Review of Literature

Kirandeep Kaur, Bharat Gurnani¹, Veena Kannusamy

Pediatric Ophthalmology and Strabismus Services, ¹Cornea and Refractive Services, Aravind Eye Hospital, Puducherry, India

Abstract

Myopia is the most common cause of refractive error in children. It is the most common ocular disorder worldwide. Apart from genetic factors, age and environmental factors have also been found to be closely associated as predictors of myopia. A comprehensive literature search was on online platforms using terms Myopia review, onset, progression, treatment, control, updates, bifocals, Atropine, and Orthokeratology. All the relevant articles published in English in last 10 years were analyzed and included. Excessive near work and prolonged screen usage have been proven as definite risk factors apart from genetics. Role of Vitamin D and outdoor activities are still having a controversial stand. Myopia treatment has come a long way from glasses/contact lenses to advanced minimally invasive refractive procedures such as femtosecond-assisted procedures and small incision lenticule extraction. With tremendous improvement in technology and increased dependence on digital devices, control of myopia progression remains a big challenge. Use of bifocals, progressive glasses, rigid contact lenses, and soft bifocals lenses have been studied widely. These all measures seem to do well in initial years, but long-standing results are not encouraging. The results with low-dose atropine have been convincing, but long-term follow-up results are still awaited.

Keywords: Atropine, myopia control, orthokeratology, progressive glasses

INTRODUCTION

Refractive error has been reported as the most common cause of reduced vision in children, affecting 2%–11% of the population below 16 years of age.^[1,2] It is also responsible for 60%–80% of visual impairment in children.^[3,4] As a single entity, myopia is the most common ocular disorder worldwide.^[5] It has been well documented that development of myopia depends both on genetic and environmental factors. In general, myopia has the trait of familial clustering. A study among Singapore Chinese preschoolers showed that family history of myopia was the strongest risk factor for offspring's myopia.^[6] Environmental risk factors such as prolonged near work, intensive education, and limited time spent outdoors are strongly supported.^[7]

Children are easily exposed to screens for prolonged times from a very young age and are dependent on tablets, smart phones, televisions, laptops, or computers. This can be related to an easy access to gadgets in today's world. Thus, there is a strong need to understand the epidemiology, etiology, associations, changing concepts, and the management options of myopia.

METHODOLOGY

In this article, we have comprehensively covered the basics as well discussed review articles published in the recent past. A thorough literature search was conducted for papers using the online search engines PubMed, Google Scholar, and Cochrane database. The following terms were used while searching articles: myopia AND (review) AND (onset OR progression OR treatment OR control OR updates OR bifocals) OR (Orthokeratology) OR (Atropine). All the relevant articles published in English language from 2010 onwards and the referenced articles were included in the study.

Myopia

Myopia is commonly referred to as short sightedness. The light rays that enter the eye are focused in front of the retina, rather than directly on it, so that distant objects appear blurred. This is illustrated in Figure 1.

Address for correspondence: Dr. Bharat Gurnani,
Aravind Eye Hospital, Puducherry, India.
E-mail: drgurnanibharat25@gmail.com

Received: 09-07-2020
Accepted: 12-07-2020

Revised: 01-09-2020
Published: 16-12-2020

Access this article online

Quick Response Code:



Website:
www.tnoajors.com

DOI:
[10.4103/tjor.tjor_85_20](https://doi.org/10.4103/tjor.tjor_85_20)

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

For reprints contact: reprints@medknow.com

How to cite this article: Kaur K, Gurnani B, Kannusamy V. Myopia: Current concepts and review of literature. TNOA J Ophthalmic Sci Res 2020;58:280-7.

The evolving epidemic of myopia

Myopia is the leading cause of visual impairment worldwide. Myopia is like an epidemic that is occurring worldwide. According to researchers, the number of myopes is expected to increase up to more than 5000 million by 2050.^[8] In the United States, the prevalence of myopia has increased from 25% to 44% between 1972 and 2004.^[9-11] In urban communities in Asia, the prevalence is >80%.^[12,13] The prevalence is much lower in underdeveloped areas of the world such as Sherpa in Nepal.^[14]

Risk factors

Age

A less hyperopic refraction at a young baseline age has been considered as the most significant predictor of myopia.^[15] In another study, it was found that for every year of delayed stabilization, there was an increase in the total amount of myopia (overall 0.27 diopters (D) more myopia per year of delay).^[16]

Hereditary

Increased incidence of myopia has been noticed in children if parents are myopic. The risk seems to be still higher if both the parents are myopic. In such scenarios, a more than six-fold increased risk of juvenile onset myopia has been reported.^[17] Parental myopia is also a risk factor for progressive myopia. Saw *et al.*^[18] showed that children with even single myopic parent had increased rates of myopia progression compared to children with no myopic parent (0.63 D/year versus 0.42 D/year, respectively).

Urbanization and increased near work

Many studies have shown an association between increased myopia in urban areas as compared to rural areas. In 2008, a Polish study found that children living in the city had two-fold increase in the rate of myopia when compared to children living in rural areas.^[19] Myopia is more common among professionals, educated patients, computer users, university students, and associated with increased intelligence. All these risk factors are associated with more near work.

Theories of myopia progression

Several theories have been proposed to explain the etiology behind myopia progression. These include:

1. Lag of accommodation
2. Mechanical tension
3. Peripheral refraction.

Lag of accommodation

The theory is based on the hypothesis that high lag of accommodation that occurs during near work in myopic eyes causes foveal hyperopic retinal blur.^[20,21] This induces an abnormal axial growth of the eye, leading to myopia as illustrated in Figure 2.

Mechanical tension

The mechanical tension theory is based on response of the eye to transient changes in axial length following short periods of

accommodation.^[22] This theory suggests that there is a forward and inward pulling of the choroid secondary to contraction of the ciliary muscle following accommodation. Such ciliary-choroidal tension restricts the equatorial growth of the eye, thus decreasing the circumference of the sclera. This leads to a more prolate shape of the eye^[22,23] and ultimately an elongation of the axial length of the eye, resulting in myopia as illustrated in Figure 3.

Peripheral refraction

Previous studies have shown that chronic exposure to lens-induced hyperopic defocus accelerates the axial growth of

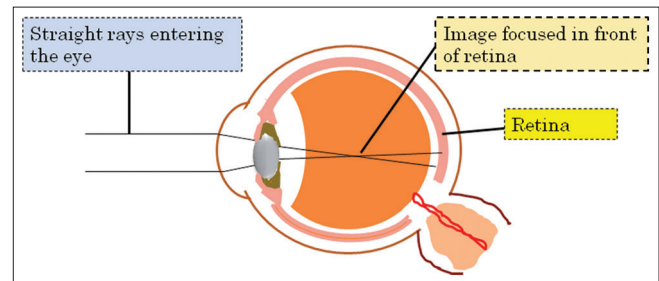


Figure 1: Parallel rays of light focus in front of the retina, simulating a myopic eye

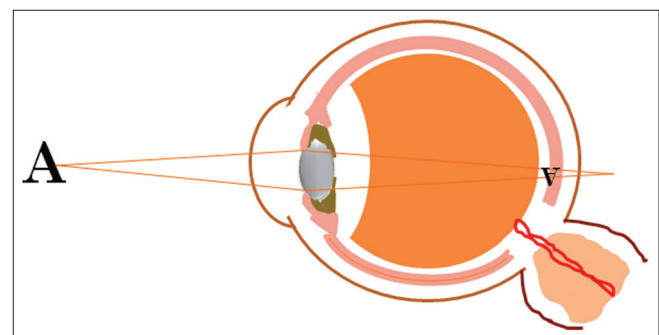


Figure 2: Foveal hyperopic retinal blur resulting from reduced accommodative response at near

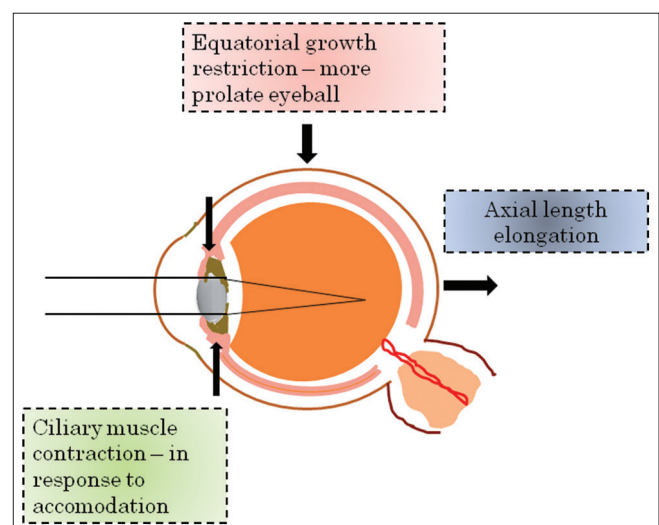


Figure 3: Schematic representation of the mechanical tension theory

the eye. Thus, it was believed that foveal defocus influences the eye growth.^[24,25] However, recent investigations on the effect of hyperopic defocus on ocular growth have highlighted the role of peripheral image formation in the etiology and progression of myopia. The peripheral refraction theory^[26,27] indicates that peripheral hyperopic defocus plays a significant role in the development of myopia as illustrated in Figure 4.

Clinical classification of myopia

Various forms of myopia have been described as:

- Congenital Myopia: It is associated with an increase in axial length and overall globe size. This is seen more frequently in children born prematurely or with birth defects, such as Marfan's or Homocystinuria
- Simple myopia: Also known as "school myopia". Myopia usually begins between 8 and 12 years of age. It is typically <4.00–6.00 diopters.^[28] This is the most common form of myopia
- Degenerative myopia: Also known as "pathological", or "progressive myopia". It is characterized by marked fundus changes, such as posterior staphyloma, and associated with a high refractive error and subnormal visual acuity after correction.^[29] This form of myopia gets progressively worse over time. This starts in childhood around 5–10 years of age and results in high myopia (>6 diopters)
- Acquired myopia: Pseudomyopia is the blurring of distance vision brought about by spasm of the accommodation system^[30]
- Nocturnal myopia: The shift from photopic to scotopic vision at twilight is associated with sensitivity to the shorter wavelengths of light. The emmetropic thus becomes slightly myopic for the shorter wavelengths^[28]
- Near work-induced transient myopia (NITM): short-term myopic far point shift immediately following a sustained near visual task.^[31] Some authors argue for a link between NITM and the development of permanent myopia^[32]
- Drug-induced myopia: It results from various medications, increases in glucose levels, oxygen toxicity (e. g., from diving or from oxygen and hyperbaric therapy) or other anomalous conditions.^[28] Sulfonamide therapy can cause ciliary body edema, resulting in anterior displacement of the lens, pushing the eye out of focus.^[33] Cholinergic drugs such as pilocarpine and echothiophate cause accommodative spasm responsible for myopia. Elevation

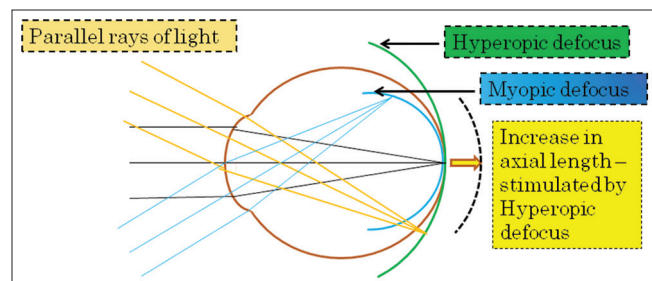


Figure 4: Schematic representation of the peripheral theory

of blood-glucose levels can also cause edema (swelling) of the crystalline lens as a result of sorbitol accumulating in the lens. This edema often causes temporary myopia

- Iatrogenic myopia: Scleral buckles, used in the repair of retinal detachments may induce myopia by increasing the axial length of the eye^[34]
- Index myopia: It is attributed to variation in the index of refraction of one or more of the ocular media.^[25] Cataracts may lead to index myopia.^[35]

Syndromic associations

The syndromic associations with myopia are listed in Table 1.

Treatment of myopia

Myopia treatment can be broadly divided into 2 subheadings as shown in Figure 5.

GENERAL TREATMENT

Optical treatment

Glasses or contact lenses

An appropriate power concave lens based on cycloplegic refraction is advised. The concave lenses diverge the light rays entering the eye and form a focused image accurately onto the retina. More severe the myopia, more stronger (more negative power) lenses are required. Basic rule while prescribing correction in myopia is minimum acceptance providing maximum vision.

However, strong eyeglass prescriptions create distortions such as prismatic movement and chromatic aberrations. Contact lens wearers do not experience these distortions because the lens moves with the cornea, keeping the optic axis in line with the visual axis. Furthermore, the vertex distance is reduced to zero with contact lenses.

Surgical treatment

The surgical management options for myopia have gained lot of importance in recent past. It is being opted by patients not only for cosmetic purpose to be able to avoid glasses but also as a means of matching the occupation vision standards. Various options for refractive surgeries for myopia are enlisted in Table 2.

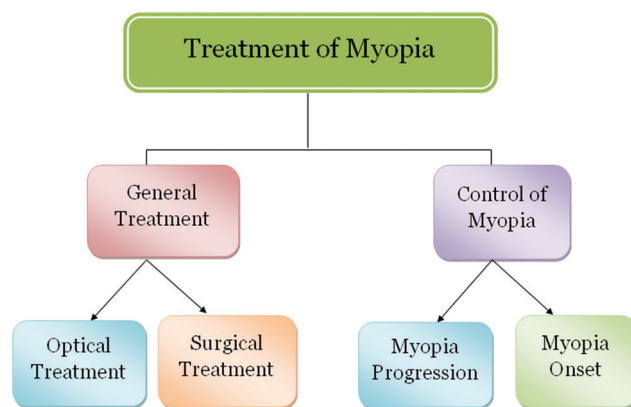


Figure 5: Diagram depicting outline of myopia management

Control of myopia

1. Control of myopia progression
2. Control of myopia onset.

Control of myopia progression

Spectacles

Historically, undercorrection of myopia was believed to slow down the progression of myopia as a result of reduced accommodation. However, with today's knowledge that blur affects the ability of the eye to become emmetropic, this has been rejected. Two recent studies have demonstrated that undercorrection actually results in mild acceleration of myopia progression.^[36,37] Thus, undercorrection should not be used to slow myopic progression.

Bifocal/multifocal glasses

Bifocal or multifocal spectacles are the most investigated for myopia control. The treatment with these glasses is based on the assumption that myopia is a response to prolonged accommodation.^[38,39] Bifocal or multifocal spectacles would reduce the accommodative effort and thus slow the progression of myopia. Cheng *et al.*^[40] studied the effect of high fitting

bifocals and base in prismatic bifocal spectacles compared to single vision (SV) glasses in myopia progression. They reported that these glasses slowed the myopia progression by 40%.

Progressive glasses

The Correction of Myopia Evaluation Trial study determined if a +2.00 D progressive additional lenses (PAL) slowed the progression of myopia as compared to SV full correcting lens.^[41] This prospective, multicenter study demonstrated that in the 1st year, PALs slowed the progression of myopia by 20%. The net reduction was 0.2 D, which was statistically significant. The PALs were most effective; when both the parents were myopic, there was a large lag of accommodation or the child had esophoria at near.^[18]

Contact lenses

For years, it was believed that gas permeable contact lenses slowed the progression of myopia. However, gas permeable contact lenses are typically prescribed when myopia begins to slow down (≥ 12 of age). It has been shown in a number of well-controlled clinical trials that neither conventional soft nor gas permeable contact lenses alter the myopia progression.^[42,43]

Soft bifocal contact lenses

The center distance (add power in the peripheral part) contact lenses have been tested for myopia control. On an average, these contact lenses slow myopia progression by 46%.^[44-47]

Lam *et al.*^[45] conducted a randomized controlled trial on 8–13 years old children with myopia between -1.00 and -5.00 spherical equivalent. Over 2 years, the myopia progressed by an average of -0.59 ± 0.49 D for the bifocal contact lens wearers and -0.79 ± 0.56 D for the SV contact lens wearers ($P = 0.03$), showing a 25% slowing of progression of myopia. Axial length elongation was also slower for the soft bifocal contact lens wearers (0.25 ± 0.23 mm) than for the SV contact lens wearers (0.37 ± 0.24 mm, $P = 0.009$).

Table 1: The common underlying ocular and syndromic associations

Syndromic associations with myopia	
Ocular diseases	Multisystem diseases
Congenital glaucoma	Stickler syndrome
Retinopathy of prematurity	Diabetes mellitus (uncontrolled)
Retinitis pigmentosa	Marfan's syndrome
Cataract	Weil-Marchesani
Congenital stationary night blindness	Knobloch
Keratoconus	Ehler danlos
Gyrate atrophy	
Albinism	

Table 2: Various surgical treatment modalities for myopia

Keratorefractive procedures		Lens-based procedures	Combined lens and cornea based refractive procedures
Incisional procedures	Radial keratotomy	Phakic refractive lenses	Bioptics
Lamellar corneal refractive procedures	Nonfreeze keratomileusis	Refractive lens exchange	Trioptics
	Keratomileusis <i>in situ</i>		
	Automated lamellar keratoplasty		
	Small incision keratoplasty		
	Corneoplastique		
Laser ablation corneal refractive procedures	PRK		
	LASEK		
	Epi-LASIK		
	LASIK		
Intracorneal implants	C-LASIK		
	Intracorneal lenses		
	Intrastromal corneal ring segments		
	Gel injectable adjustable keratoplast		

PRK: Photorefractive keratectomy, LASEK: Laser subepithelial keratomileusis, Epi-LASIK: Epithelial laser *in situ* keratomileusis, C-LASIK: Custom laser *in situ* keratomileusis

Orthokeratology lenses

Orthokeratology (OK) lenses have been a boon for myopes. It provides patients with a “wow” factor and eliminates the need of daily wearing of contact lenses or glasses. This is particularly beneficial for athletes. Majority achieve the visual acuity of 20/20 and over 90% achieve 20/30.^[48]

OK lenses change the curvature of the cornea by mechanical flattening of the cornea as shown in Figure 6. However, there is a strong evidence that the change in refraction is achieved by horizontal movement of epithelial cells secondary to reverse pressure made from the seal created in the mid-periphery bearing area of the lens.^[49,50]

Reim *et al.*^[51] performed a retrospective study of 253 children (ages 6–18) on the ability of OK to slow the progression of myopia. They reported that the rate of progression was slowed from 0.5 to 0.13 D/year. Subsequently, there have been a number of prospective clinical trials, which have demonstrated that OK lenses slows the progression of myopia by 40% using axial length measurements and wash-out cycloplegic measurements.^[52-60]

Pharmaceutical agents- Antimuscarinic agents

Atropine was first used by Wells in 1900 to stop the progression of myopia by “paralyzing” accommodation. Analysis of a number of retrospective studies using atropine has shown that 1% atropine tends to slow the progression of myopia by almost 80%.

Atropine is a nonspecific muscarinic receptor antagonist that causes cycloplegia and mydriasis, and pirenzepine is a M1-specific muscarinic receptor antagonist. M1 receptors are highly concentrated in the retina and found rarely on the ciliary body or iris, so they cause far fewer cycloplegic and mydriasis-related symptoms. Both pirenzepine^[61,62] and atropine^[63-65] have been shown to slow myopia progression. The exact mechanism by which atropine inhibits myopia progression is unknown. Multiple studies indicate that atropine has an effect altering the sclera.^[66-68] It has also been suggested that ultraviolet light (secondary to pupil dilation) may increase collagen cross-linking within the sclera, thereby slowing down the scleral growth.

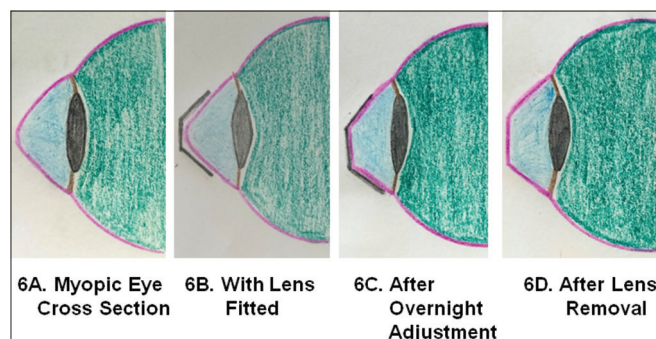


Figure 6: Schematic diagram showing effect of overnight Orthokeratology lenses

Chua *et al.*^[64] (ATOM 1) studied the effect of 1% atropine in a group of 400 children (13.5% dropout rate). One group received atropine, whereas the other group received a placebo. Only one eye of each child was chosen for treatment. The mean progression in the control eye after 2 years was 0.6 D/year and in the atropine-treated eye was 0.14 D/year. This represents a 77% reduction in the progression of myopia. Furthermore, the axial length measurements in the eyes, which received atropine, remained almost unchanged (0.02 mm over 2 years). There were no serious adverse events and the atropine was well tolerated.

Chia *et al.*^[69] (ATOM 2) studied the effect of 0.5%, 0.1%, and 0.01% concentrations of atropine in a group of 200 children. After 2 years, all 3 concentrations slowed myopia progression. Mean progression with each concentration (spherical equivalent) was 0.15 D/year (0.5% atropine), 0.19 D/year (0.1% atropine), and 0.25 D/year (0.01% atropine). ATOM 2 study suggests that myopic progression was slowed with all concentrations, with similar effects between moderate and low concentrations.

In contrast to the benefits, the primary ocular side effects of topical atropine include mydriasis leading to photophobia, loss of accommodation resulting in blurred near vision, and local allergic responses. In addition to the side effects, high concentration atropine leads to a significant rebound following cessation of eye drops.^[70]

Prevention of myopia onset

Limit screen time

The increased screen exposure could be associated with a higher risk of preschool myopia. It is proposed that the closer the viewing distance, more the focusing errors^[71,72] and higher the lag of accommodation.^[71,73]

Studies concerning screen exposure and myopia have been conducted worldwide. A study involving primary and middle school students in six provinces of China showed that children had a higher risk of myopia whose parents did not limit their offspring's screen time;^[74] Harrington *et al.* reported that using screens >3 h/day was associated with a higher risk of myopia among schoolchildren in Ireland.^[75] Similarly, a study from North India found screen viewing was a significant risk factor for myopia progression amongst children aged 5–15 years.^[76]

The current American Academy of Pediatrics guidelines recommend that children under 2 years of age should not spend any time using electronic media, while children over 2 years of age should be restricted to <2 h/day.^[77,78]

Increased outdoor time

Jones *et al.*^[79] first reported the association between outdoor time and likelihood of developing myopic refractive error. This similar effect has been reported in several subsequent studies.^[80-85]

The indoor activities create more hyperopic defocus (causing myopia) across the entire surface of the retina than any

outdoors activities. Outdoor activities essentially eliminate any defocus across the entire visual field that serves as a stop signal for the eye growth (thus inhibiting development of myopia). Brighter light intensity also leads to pupil constriction and increased depth of focus, which reduces optical blur and increases contrast. Change in contrast, in turn, would affect the function of amacrine cells, which might explain the role of dopamine in myopia development in animal models. Few studies have tried to find an association between levels of Vitamin D, time spent outdoors, and myopia. The results have not been consistent, with some favoring and disapproving this association.^[86,87] Spending more time outdoors clearly has a substantial therapeutic effect on myopia onset and possibly progression. Therefore, it should be recommended that children, especially those who have two myopic parents or show signs of myopia development or progression, spend more time outdoors as preventive measure of developing myopia.

Low concentration atropine

Chia *et al.*^[88] conducted a study in which children between the ages of 6 and 12 years with +1.00 D and -1.00 D spherical equivalent, cycloplegic refractive error were followed for at least 12 months and included in a retrospective comparison of children who received 0.025% atropine and those who did not. Only 21% of the children receiving atropine became myopic, compared with 54% of the children not receiving atropine ($P = 0.016$). Refractive error progression was also less for those on atropine (-0.14 ± 0.24 D/year) than for those not on atropine (-0.58 ± 0.34 D/year, $P = 0.0001$). No children in either group complained of blurry vision at near, and there was no difference in reports of photophobia for those on atropine (16%) and those not on atropine (8%, $P = 0.41$).

For clinical practice, it should be remembered that atropine 0.01% eye drops are probably the most useful current way to reduce progression, but that this is a non-reimbursable, off-label treatment in Germany. The age up to which atropine eye drop treatment should be continued, the optimal length of treatment, and the nature of progression after the end of treatment are unknown. It also remains unproven whether atropine treatment may be useful as a prophylactic measure to prevent subsequent myopia, as has been shown for exposure to daylight.

Future prospects

The prevalence of myopia worldwide is increasing exponentially. The epidemic waves are seen in developing countries like India as well. Also, there is markedly increased rates of progression to pathological myopia. There is a need to understand the factors responsible for myopic waves in each country individually. Clinical trials to control myopia progression are needed on large scale. A comprehensive analysis of risk factors, lifestyle modifications to prevent myopia, and analysis of different modalities to prevent progression to pathological myopia once it sets in will help in better understanding the disease. These will also help lawmakers to ensure that school curriculums and teachings

are modified in a way to control the myopia explosion. Family-based approach needs to be adapted as it would be ideal to understand the individual risk factors. A proactive approach prior to onset of myopia will be beneficial. Earlier implementation of interventions will also help significantly reduce the chance of progression to pathological myopia.

CONCLUSION

Excessive near work and prolonged screen usage have been proven as definite risk factors apart from genetics. Role of Vitamin D and outdoor activities are still having a controversial stand. Myopia treatment has come a long way from glasses/contact lenses to advanced minimally invasive refractive procedures such as femtosecond assisted procedures and small incision lenticule extraction.

With tremendous improvement in technology and increased dependence on digital devices, control on progression of myopia remains a big challenge. Use of bifocals, progressive glasses, rigid contact lenses, and soft bifocals lenses have been studied widely. These all measures seem to do well in initial years, but long-standing results are not encouraging. Use of low-dose atropine has been approved, but long-term results are still awaited.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

REFERENCES

1. Padhye AS, Khandekar R, Dharmadhikari S, Dole K, Gogate P, Deshpande M. Prevalence of uncorrected refractive error and other eye problems among urban and rural school children. *Middle East Afr J Ophthalmol* 2009;16:69-74.
2. Dirani M, Chan YH, Gazzard G, Hornbeak DM, Leo SW, Selvaraj P, *et al.* Prevalence of refractive error in Singaporean Chinese children: The strabismus, amblyopia, and refractive error in young Singaporean Children (STARS) study. *Invest Ophthalmol Vis Sci* 2010;51:1348-55.
3. Dandona R, Dandona L, Srinivas M, Sahare P, Narsaiah S, Muñoz SR, *et al.* Refractive error in children in a rural population in India. *Invest Ophthalmol Vis Sci* 2002;43:615-22.
4. Murthy GV, Gupta SK, Ellwein LB, Muñoz SR, Pokharel GP, Sanga L, *et al.* Refractive error in children in an urban population in New Delhi. *Invest Ophthalmol Vis Sci* 2002;43:623-31.
5. Pararajasegaram R. VISION 2020-the right to sight: From strategies to action. *Am J Ophthalmol* 1999;128:359-60.
6. Low W, Dirani M, Gazzard G, Chan YH, Zhou HJ, Selvaraj P, *et al.* Family history, near work, outdoor activity, and myopia in Singapore Chinese preschool children. *Br J Ophthalmol* 2010;94:1012-6.
7. Morgan IG, Ohno-Matsui K, Saw SM. Myopia. *Lancet* 2012;379:1739-48.
8. Holden BA, Fricke TR, Wilson DA, Jong M, Naidoo KS, Sankaridurg P, *et al.* Global prevalence of myopia and high myopia and temporal trends from 2000 through 2050. *Ophthalmology* 2016;123:1036-42.
9. Kempen JH, Mitchell P, Lee KE, Tielsch JM, Broman AT, Taylor HR, *et al.* The prevalence of refractive errors among adults in the United States, Western Europe, and Australia. *Arch Ophthalmol* 2004;122:495-505.
10. Javitt JC, Chiang YP. The socioeconomic aspects of laser refractive surgery. *Arch Ophthalmol* 1994;112:1526-30.

11. Vitale S, Sperduto RD, Ferris FL 3rd. Increased prevalence of myopia in the United States between 1971-1972 and 1999-2004. *Arch Ophthalmol* 2009;127:1632-9.
12. Lin LL, Shih YF, Hsiao CK, Chen CJ. Prevalence of myopia in Taiwanese schoolchildren: 1983 to 2000. *Ann Acad Med Singapore* 2004;33:27-33.
13. Lam CS, Goldschmidt E, Edwards MH. Prevalence of myopia in local and international schools in Hong Kong. *Optom Vis Sci* 2004;81:317-22.
14. Niroula DR, Saha CG. Study on the refractive errors of school going children of Pokhara city in Nepal. *Kathmandu Univ Med J* 2009;7:67-72.
15. French AN, Morgan IG, Mitchell P, Rose KA. Risk factors for incident myopia in Australian schoolchildren: The Sydney adolescent vascular and eye study. *Ophthalmology* 2013;120:2100-8.
16. Group COMET. Myopia stabilization and associated factors among participants in the correction of myopia evaluation trial. (COMET) *Invest Ophthalmol Vis Sci* 2013;54:7871-84.
17. Pacella R, McLellan J, Grice K, Del Bono EA, Wiggs JL, Gwiazda JE. Role of genetic factors in the etiology of juvenile-onset myopia based on a longitudinal study of refractive error. *Optom Vis Sci* 1999;76:381-6.
18. Saw SM, Nieto FJ, Katz J, Schein OD, Levy B, Chew SJ. Familial clustering and myopia progression in Singapore school children. *Ophthalmic Epidemiol* 2001;8:227-36.
19. Czepita D, Mojsa A, Zejmo M. Prevalence of myopia and hyperopia among urban and rural schoolchildren in Poland. *Ann Acad Med Stetin* 2008;54:17-21.
20. Gwiazda JI, Thorn F, Bauer J, Held R. Myopic children show insufficient accommodative response to blur. *Invest Ophthalmol Vis Sci* 1993;34:690-4.
21. López-Gil N, Martin J, Liu T, Bradley A, Díaz-Muñoz D, Thibos LN. Retinal image quality during accommodation. *Ophthalmic Physiol Opt* 2013;33:497-507.
22. Drexler W, Findl O, Schmetterer L, Hitzinger CK, Fercher AF. Eye elongation during accommodation in humans: Differences between emmetropes and myopes. *Invest Ophthalmol Vis Sci* 1998;39:2140-7.
23. Mutti DO, Sholtz RI, Friedman NE, Zadnik K. Peripheral refraction and ocular shape in children. *Invest Ophthalmol Vis Sci* 2000;41:1022-30.
24. Schaeffel F, Glasser A, Howland HC. Accommodation, refractive error and eye growth in chickens. *Vision Res* 1988;28:639-57.
25. Hung LF, Crawford ML, Smith EL. Spectacle lenses alter eye growth and the refractive status of young monkeys. *Nat Med* 1995;1:761-5.
26. Smith EL 3rd, Kee CS, Ramamirtham R, Qiao-Grider Y, Hung LF. Peripheral vision can influence eye growth and refractive development in infant monkeys. *Invest Ophthalmol Vis Sci* 2005;46:3965-72.
27. Smith EL 3rd, Ramamirtham R, Qiao Grider Y, Hung LF, Huang J, Kee CS, *et al.* Effects of foveal ablation on emmetropization and form deprivation myopia. *Invest Ophthalmol Vis Sci* 2007;48:3914-22.
28. American Optometric Association. Optometric Clinical Practice Guideline: Care of the Patient with Myopia; 22 January, 2015.
29. Cline D, Hofstetter HW, Griffin JR. *Dictionary of Visual Science*. 4th ed.. Boston: Butterworth-Heinemann; 1997.
30. Cassin B, Solomon S. *Dictionary of Eye Terminology*. Gainesville, Florida: Triad Publishing Company; 2001.
31. Ong E, Ciuffreda KJ. Nearwork-induced transient myopia: A critical review. *Doc Ophthalmol* 1995;91:57-85.
32. Ciuffreda KJ, Vasudevan B. Nearwork-induced transient myopia (NITM) and permanent myopia-is there a link? *Ophthalmic Physiol Optics* 2008;28:103-14.
33. Panday VA, Rhee DJ. Review of sulfonamide-induced acute myopia and acute bilateral angle-closure glaucoma. *Compr Ophthalmol Update* 2007;8:271-6.
34. Vukojević N, Sikić J, Curković T, Juratovac Z, Katusić D, Sarić B, *et al.* Axial eye length after retinal detachment surgery. *Coll Antropol* 2005;29 Suppl 1:25-7.
35. Metge P, Donnadiou M. Myopia and cataract. *La Revue du Praticien (in French)* 1993;43:1784-6.
36. Chung K, Mohidin N, O'Leary DJ. Undercorrection of myopia enhances rather than inhibits myopia progression. *Vision Res* 2002;42:2555-9.
37. Adler D, Millodot M. The possible effect of undercorrection on myopic progression in children. *Clin Exp Optom* 2006;89:315-21.
38. Rosenfield M, Gilmartin B. Accommodative error, adaptation and myopia. *Ophthalmic Physiol Opt* 1999;19:159-64.
39. Gwiazda J, Thorn F, Held R. Accommodation, accommodative convergence, and response AC/A ratios before and at the onset of myopia in children. *Optom Vis Sci* 2005;82:273-8.
40. Cheng D, Schmid KL, Woo GC, Drobe B. Randomized trial of effect of bifocal and prismatic bifocal spectacles on myopic progression: Two-year results. *Arch Ophthalmol* 2010;128:12-9.
41. Gwiazda JE, Hyman L, Everett D, Norton T, Kurtz D, Manny R., *et al.* Five year results from the correction of myopia evaluation trial (COMET). *Invest Ophthalmol Vis Sci* 2006;47:1166.
42. Walline JJ, Jones LA, Mutti DO, Zadnik K. A randomized trial of the effects of rigid contact lenses on myopia progression. *Arch Ophthalmol* 2004;122:1760-6.
43. Walline JJ, Jones LA, Sinnott L, Manny RE, Gaume A, Rah MJ, *et al.* A randomized trial of the effect of soft contact lenses on myopia progression in children. *Invest Ophthalmol Vis Sci* 2008;49:4702-6.
44. Anstice NS, Phillips JR. Effect of dual-focus soft contact lens wear on axial myopia progression in children. *Ophthalmology* 2011;118:1152-61.
45. Lam CS, Tang WC, Tse DY, Tang YY, To CH. Defocus incorporated soft contact (DISC) lens slows myopia progression in Hong Kong Chinese schoolchildren: A 2-year randomised clinical trial. *Br J Ophthalmol* 2014;98:40-5.
46. Walline JJ, Greiner KL, McVey ME, Jones-Jordan LA. Multifocal contact lens myopia control. *Optom Vis Sci* 2013;90:1207-14.
47. Sankaridurg P, Holden B, Smith E 3rd, Naduvilath T, Chen X, de la Jara PL, *et al.* Decrease in rate of myopia progression with a contact lens designed to reduce relative peripheral hyperopia: One-year results. *Invest Ophthalmol Vis Sci* 2011;52:9362-7.
48. Rah MJ, Jackson JM, Jones LA, Marsden HJ, Bailey MD, Barr JT. Overnight orthokeratology: Preliminary results of the lenses and overnight orthokeratology (LOOK) study. *Optom Vis Sci* 2002;79:598-605.
49. Zhong X, Chen X, Xie RZ, Yang J, Li S, Yang X, *et al.* Differences between overnight and longterm wear of orthokeratology contact lenses in corneal contour, thickness, and cell density. *Cornea* 2009;28:271-9.
50. Yeh TN, Green HM, Zhou Y, Pitts J, Kitamata-Wong B, Lee S, *et al.* Short-term effects of overnight orthokeratology on corneal epithelial permeability and biomechanical properties. *Invest Ophthalmol Vis Sci* 2013;54:3902-11.
51. Reim T, Lund M, Wu R. Orthokeratology and adolescent myopia control. *Contact Lens Spectr* 2003;18:40-2.
52. Lui WO, Edwards MH. Orthokeratology in low myopia. Part I: Efficacy and predictability. *Cont Lens Anterior Eye* 2000;23:77-89.
53. Walline JJ, Rah MJ, Jones LA. The Children's overnight orthokeratology investigation (COOKI) pilot study. *Optom Vis Sci* 2004;81:407-13.
54. Cho P, Cheung SW, Edwards M. The longitudinal orthokeratology research in children (LORIC) in Hong Kong: A pilot study on refractive changes and myopic control. *Curr Eye Res* 2005;30:71-80.
55. Walline JJ, Jones LA, Sinnott LT. Corneal reshaping and myopia progression. *Br J Ophthalmol* 2009;93:1181-5.
56. Kakita T, Hiraoka T, Oshika T. Influence of overnight orthokeratology on axial elongation in childhood myopia. *Invest Ophthalmol Vis Sci* 2011;52:2170-4.
57. Santodomingo-Rubido J, Villa-Collar C, Gilmartin B, Gutierrez-Ortega R. Myopia control with orthokeratology contact lenses in Spain (MCOS): Refractive and Biometric changes. *Invest Ophthalmol Vis Sci* 2012;53:5060-5.
58. Swarbrick HA, Alharbi A, Watt K, Lum E, Kang P. Myopia control during orthokeratology lens wear in children using a novel study design. *Ophthalmology* 2015;122:620-30.
59. Kwok-Hei Mok A, Sin-Ting Chung C. Seven-year retrospective analysis of the myopic control effect of orthokeratology in children: A pilot study. *Clin Optom* 2011;3:1-4.
60. Hiraoka T, Kakita T, Okamoto F, Takahashi H, Oshika T. Long-term effect of overnight orthokeratology on axial length elongation in childhood myopia: A 5-year follow-up study. *Invest Ophthalmol Vis Sci* 2012;53:3913-9.
61. Siatkowski RM, Cotter SA, Crockett RS, Miller JM, Novack GD, Zadnik K, *et al.* Two-year multicenter, randomized, double-masked, placebo-controlled, parallel safety and efficacy study of 2% pirenzepine

- ophthalmic gel in children with myopia. *J AAPOS* 2008;12:332-9.
62. Tan DT, Lam DS, Chua WH, Shu-Ping DF, Crockett RS; Asian Pirenzepine Study Group. One-year multicenter, double-masked, placebo-controlled, parallel safety and efficacy study of 2% pirenzepine ophthalmic gel in children with myopia. *Ophthalmology* 2005;112:84-91.
63. Wu PC, Yang YH, Fang PC. The long-term results of using lowconcentration atropine eye drops for controlling myopia progression in schoolchildren. *J Ocul Pharmacol Ther* 2011;27:461-6.
64. Chua WH, Balakrishnan V, Chan YH, Tong L, Ling Y, Quah BL, *et al.* Atropine for the treatment of childhood myopia. *Ophthalmology* 2006;113:2285-91.
65. Lee JJ, Fang PC, Yang IH, Chen CH, Lin PW, Lin SA, *et al.* Prevention of myopia progression with 0.05% atropine solution. *J Ocul Pharmacol Ther* 2006;22:41-6.
66. Zou L, Liu R, Zhang X, Chu R, Dai J, Zhou H, *et al.* Upregulation of regulator of G-protein signaling 2 in the sclera of a form deprivation myopic animal model. *Mol Vis* 2014;20:977-87.
67. Gallego P, Martínez-García C, Pérez-Merino P, Ibares-Frías L, Mayo-Iscar A, Merayo-Llodes J. Scleral changes induced by atropine in chicks as an experimental model of myopia. *Ophthalmic Physiol Opt* 2012;32:478-84.
68. Barathi VA, Beuerman RW. Molecular mechanisms of muscarinic receptors in mouse scleral fibroblasts: Prior to and after induction of experimental myopia with atropine treatment. *Mol Vis* 2011;17:680-92.
69. Chia A, Chua WH, Cheung YB, Wong WL, Lingham A, Fong A, *et al.* Atropine for the treatment of childhood myopia: Safety and efficacy of 0.5%, 0.1%, and 0.01% doses (Atropine for the Treatment of Myopia 2). *Ophthalmology* 2012;119:347-54.
70. Tong L, Huang XL, Koh AL, Zhang X, Tan DT, Chua WH. Atropine for the treatment of childhood myopia: Effect on myopia progression after cessation of atropine. *Ophthalmology* 2009;116:572-9.
71. Harb E, Thorn F, Troilo D. Characteristics of accommodative behavior during sustained reading in emmetropes and myopes. *Vis Res* 2006;46:2581-92.
72. Abbott ML, Schmid KL, Strang NC. Differences in the accommodation stimulus response curves of adult myopes and emmetropes. *Ophthalmic Physiol Opt* 1998;18:13-20.
73. Berntsen DA, Sinnott LT, Mutti DO, Zadnik K; CLEERE Study Group. Accommodative lag and juvenile-onset myopia progression in children wearing refractive correction. *Vision Res* 2011;51:1039-46.
74. Zhou J, Ma Y, Ma J, Zou Z, Meng X, Tao F, *et al.* Prevalence of myopia and influencing factors among primary and middle school students in 6 provinces of China. *Zhonghua Liu Xing Bing Xue Za Zhi* 2016;37:29-34.
75. Harrington SC, Stack J, O'Dwyer V. Risk factors associated with myopia in schoolchildren in Ireland. *Br J Ophthalmol* 2019;103:1803-9.
76. Saxena R, Vashist P, Tandon R, Pandey RM, Bhardawaj A, Gupta V, *et al.* Incidence and progression of myopia and associated factors in urban school children in Delhi: The North India Myopia Study (NIM Study). *PLoS One* 2017;12:e0189774.
77. American Academy of Pediatrics. Policy statement: Children, adolescents, and the media. *Pediatrics* 2013;132:958-61.
78. Council on Communications and Media, Brown A. Media use by children younger than 2 years. *Pediatrics* 2011;128:1040-5.
79. Jones LA, Sinnott LT, Mutti DO, Mitchell GL, Moeschberger ML, Zadnik K. Parental history of myopia, sports and outdoor activities, and future myopia. *Invest Ophthalmol Vis Sci* 2007;48:3524-32.
80. Rose KA, Morgan IG, Ip J, Kifley A, Huynh S, Smith W, *et al.* Outdoor activity reduces the prevalence of myopia in children. *Ophthalmology* 2008;115:1279-85.
81. Dirani M, Tong L, Gazzard G, Zhang X, Chia A, Young TL, *et al.* Outdoor activity and myopia in Singapore teenage children. *Br J Ophthalmol* 2009;93:997-1000.
82. Guggenheim JA, Northstone K, McMahon G, Ness AR, Deere K, Mattocks C, *et al.* Time outdoors and physical activity as predictors of incident myopia in childhood: A prospective cohort study. *Invest Ophthalmol Vis Sci* 2012;53:2856-65.
83. Guo Y, Liu LJ, Xu L, Lv YY, Tang P, Feng Y, *et al.* Outdoor activity and myopia among primary students in rural and urban regions of Beijing. *Ophthalmology* 2013;120:277-83.
84. Lin Z, Vasudevan B, Jhanji V, Mao GY, Gao TY, Wang FH, *et al.* Near work, outdoor activity, and their association with refractive error. *Optom Vis Sci* 2014;91:376-82.
85. Pan CW, Qian DJ, Saw SM. Time outdoors, blood vitamin D status and myopia: A review. *Photochem Photobiol Sci* 2017;16:426-32.
86. Tideman JW, Polling JR, Voortman T, Jaddoe VW, Uitterlinden AG, Hofman A, *et al.* Low serum vitamin D is associated with axial length and risk of myopia in young children. *Eur J Epidemiol* 2016;31:491-9.
87. Wu PC, Tsai CL, Wu HL, Yang YH, Kuo HK. Outdoor activity during class recess reduces myopia onset and progression in school children. *Ophthalmology* 2013;120:1080-5.
88. Fang PC, Chung MY, Yu HJ, Wu PC. Prevention of myopia onset with 0.025% atropine in premyopic children. *J Ocul Pharmacol Ther* 2010;26:341-5.